Risk-of-bias tool chooser

Which appraisal tool for which study design?

Foundations of Clinical Research · Session 4 (4.3) and Session 8 (8.3)

How to choose

- Step 1: identify the design of the study (use the design decision tree from Session 3).
- Step 2: pick the tool in the table. Appraise each outcome you care about, not the whole paper at once.
- Step 3: judge each domain (low / some concerns / high risk), then ask: are the conclusions supported by the data? In the course we use the simplified 10-question checklist; use these full tools when you appraise evidence formally (e.g. for a systematic review).
- Reporting guidelines (CONSORT, STROBE, ...) tell authors what to REPORT. Risk-of-bias tools judge how TRUSTWORTHY the result is. They are not the same thing.

Study design	Tool	What it checks	Free source
Randomised controlled trial	**RoB 2**	Bias from randomisation, deviations from intended interventions, missing outcome data, outcome measurement and selection of the reported result	Cochrane, riskofbias.info
Non-randomised study of an intervention	**ROBINS-I**	Confounding, selection of participants, classification of interventions, deviations, missing data, measurement and reporting, judged against an ideal “target trial”	Cochrane, riskofbias.info
Cohort and case-control (exposures)	**Newcastle–Ottawa Scale** or **ROBINS-E**	NOS: stars for selection of groups, comparability (confounding) and outcome/exposure ascertainment. ROBINS-E: the ROBINS-I domains, for exposures	Ottawa Hospital Research Institute, ohri.ca; riskofbias.info
Diagnostic accuracy study	**QUADAS-2**	Patient selection, index test, reference standard, flow and timing: risk of bias AND applicability	University of Bristol, bristol.ac.uk (QUADAS)
Systematic review / meta-analysis	**AMSTAR 2**	16 items: protocol registered in advance, comprehensive search, risk-of-bias assessment of included studies, appropriate meta-analysis, publication bias	amstar.ca
Qualitative study	**CASP Qualitative Checklist**	10 questions: clear aims, appropriate method and design, recruitment, data collection, researcher relationship (reflexivity), ethics, rigour of analysis, findings, value	Critical Appraisal Skills Programme, casp-uk.net

Other designs: PROBAST for prediction models, and the JBI critical appraisal checklists (jbi.global) for cross-sectional studies and case series. All tools above are free to download; check the source for the latest version and its guidance document.